CHAPTER 5

CELL: THE UNIT OF LIFE

CHAPTER ESSENCE AT A GLANCE:
The fundamental structural and functional unit of life characterized by membrane compartmentalization and macromolecular assemblies.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Cell Theory and Historical Foundations:** Schleiden and Schwann established that all living organisms are composed of cells and cell products. Rudolf Virchow (1855) completed cell theory with 'Omnis cellula-e cellula', establishing that all new cells arise strictly from the division of pre-existing cells.

• **Fluid Mosaic Model of Biomembranes:** Singer and Nicolson (1972) proposed the Fluid Mosaic Model of the plasma membrane: a quasi-fluid phospholipid bilayer with hydrophilic polar phosphate heads directed outward and hydrophobic fatty acid tails sequestered inward, populated by mobile integral and peripheral proteins allowing selective permeability.

• **The Endomembrane System:** The endomembrane system coordinates organelle functions through vesicle trafficking: Rough Endoplasmic Reticulum (RER with ribosomes synthesizing proteins), Smooth Endoplasmic Reticulum (SER synthesizing lipids and steroid hormones), Golgi apparatus (cis-to-trans glycosylation and packaging into transport vesicles), Lysosomes (acid hydrolases), and Vacuoles.

• **Semi-Autonomous Organelles and Cytoskeleton:** Mitochondria and Chloroplasts are semi-autonomous double-membraned organelles possessing circular double-stranded DNA, 70S ribosomes, and self-dividing ability via binary fission. Eukaryotic cilia and flagella possess an axoneme with a 9+2 microtubular doublet arrangement anchored to basal bodies.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{Endomembrane Flow} = \text{RER and SER} \xrightarrow{\text{Vesicles}} \text{Golgi cis} \rightarrow \text{trans} \rightarrow \text{Lysosome and Vacuole}$$

Functional Coordination of the Endomembrane System

$$\text{Axoneme Microtubules} = 9 \text{ Peripheral Doublets} + 2 \text{ Central Singlets} \quad (9+2 \text{ Pattern})$$

Ciliary and Flagellar Cytoskeletal Architecture

$$\text{Semi-Autonomous Organelles} = \text{Double Membrane} + \text{Circular DNA} + \text{70S Ribosomes}$$

Mitochondrial and Plastid Endosymbiotic Markers

$$\text{Chromosome Classification} = \frac{\text{Metacentric (Middle)}}{\text{Sub-metacentric (Sub-middle)}} + \frac{\text{Acrocentric (Near End)}}{\text{Telocentric (Terminal)}}$$

Centromere Position Morphologies in Chromosomes

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Mitochondria, Chloroplasts, and Peroxisomes are NOT considered part of the endomembrane system because their functions are not coordinated with ER, Golgi, and lysosomes.

• **Alert:** Lysosomal hydrolytic enzymes (lipases, proteases, carbohydrases) function optimally at an acidic pH (~5.0) maintained by active proton pumping.

• **Alert:** Ribosomes are non-membrane bound organelles found in both prokaryotes (70S) and eukaryotes (80S in cytoplasm, 70S in mitochondria and chloroplasts).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State the classical cell theory formulated by Schleiden and Schwann. What crucial modification was introduced by Rudolf Virchow? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Schleiden and Schwann formulated: (1) All living organisms are composed of cells and products of cells, and (2) The cell is the fundamental structural unit of life. **[1 Mark]**

• Rudolf Virchow added the generalization 'Omnis cellula-e cellula' (1855), explaining that all cells arise strictly from the division of pre-existing cells. **[1 Mark]**

2. Why are lysosomes termed the suicidal bags of the cell? State the optimal pH range for their enzymatic activity. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Lysosomes are membrane-bound vesicular structures containing concentrated hydrolytic digestive enzymes that, if ruptured, digest and destroy the cell's own autologous components. **[1 Mark]**

• Their hydrolytic enzymes (hydrolases) function optimally at an acidic pH (~4.5 to 5.0). **[1 Mark]**

3. Describe the Fluid Mosaic Model of the plasma membrane proposed by Singer and Nicolson, detailing the arrangement of lipids and proteins. [Score: 3 Marks]

Answer & Key Valuation Steps:

• The model depicts a phospholipid bilayer in which lipids exhibit a quasi-fluid state that enables lateral movement of proteins within the overall bilayer. **[1 Mark]**

• Phospholipids have hydrophilic polar heads facing outward toward aqueous environments and hydrophobic fatty acid tails sequestered inward away from water. **[1 Mark]**

• Proteins are classified into peripheral proteins lying on the membrane surface and integral proteins partially or totally buried across the lipid bilayer. **[1 Mark]**

4. Name the four organelles included in the endomembrane system. Explain why mitochondria and chloroplasts are excluded from this group. [Score: 3 Marks]

Answer & Key Valuation Steps:

• The endomembrane system includes: (1) Endoplasmic Reticulum, (2) Golgi apparatus, (3) Lysosomes, and (4) Vacuoles. **[1.5 Marks]**

• Mitochondria, chloroplasts, and peroxisomes are excluded because their functions (cellular respiration, photosynthesis, peroxide degradation) are independent and not coordinated with the endomembrane trafficking system. **[1.5 Marks]**

5. Explain the 9+2 internal microtubular arrangement of eukaryotic cilia and flagella. [Score: 3 Marks]

Answer & Key Valuation Steps:

• The core of eukaryotic cilia and flagella is termed the axoneme, enclosed by the plasma membrane. **[1 Mark]**

• The axoneme possesses nine evenly spaced pairs of peripheral doublets of microtubules arranged radially around the periphery. **[1 Mark]**

• A single pair of centrally located singlets is connected by a bridge and enclosed by a central sheath, linked to peripheral doublets by radial spokes, constituting the 9+2 array. **[1 Mark]**

6. Classify chromosomes into four morphological categories based on the position of the centromere, describing the relative arm lengths of each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Metacentric: Centromere lies in the middle, forming two equal arms of the chromosome. **[0.75 Mark]**

• Sub-metacentric: Centromere is situated slightly away from the middle, resulting in one slightly shorter arm and one longer arm. **[0.75 Mark]**

• Acrocentric: Centromere is situated close to its end, forming one extremely short arm and one very long arm. **[0.75 Mark]**

• Telocentric: Centromere is terminal, situated at the extreme tip of the chromosome. **[0.75 Mark]**

7. Why are Mitochondria and Chloroplasts termed semi-autonomous organelles? Justify with four pieces of structural and functional evidence. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Own genetic material: Both organelles possess their own circular, double-stranded DNA molecules distinct from nuclear DNA. **[1 Mark]**

• Protein synthesis machinery: Both possess their own 70S ribosomes, RNA molecules, and enzymes to synthesize several of their own structural proteins. **[1 Mark]**

• Self-duplication: They reproduce independently within the cell by binary fission similar to prokaryotic bacteria. **[1 Mark]**

• Double membrane envelope: Outer membrane is porous, while inner membrane is selectively permeable, hosting energy-transducing ATP synthase complexes. **[1 Mark]**